

Week 1 — Deep Implementation Guide

AI-Powered Channel Partner & Lead Intelligence Dashboard

Intern	Ansh Rohilla	Period	1–7 June 2026
Organization	Vyana Innovations Pvt. Ltd.	Phase	Week 1 of 5
Mentor	Anshika Rohilla, HR	Document	Implementation Guide v1.0

This document provides a complete, day-by-day implementation reference for Week 1 of the internship. Every day includes exact terminal commands to run, code files to create, and deliverables to produce. Follow this guide sequentially — by end of Day 7 you will have a fully scaffolded project, a requirements document, a real sample dataset, a system architecture, and complete wireframes ready to present to your mentor.

Week 1 at a Glance

Day	Focus	Key Deliverable
Day 1	Onboarding & environment setup	Dev environment verified, git repo initialized
Day 2	Requirements gathering & data schema	requirements_v1.md + data_schema.md
Day 3	Sample dataset creation	partners.csv (80 rows), leads.csv (400 rows)
Day 4	System architecture design	Architecture diagram + API endpoint spec
Day 5	Tech stack & sprint board	requirements.txt + package.json + sprint board live
Day 6	UI/UX wireframing	5-page wireframe doc (Excalidraw/Figma export)
Day 7	Week 1 review & mentor check-in	Week 1 summary + revised roadmap

Day 1

Onboarding & Environment Setup

1 June 2026

Goal: Leave Day 1 with every tool installed, the git repo initialized, and all Python packages verified working. Don't skip the verification step at the end.

Morning — Company Orientation Checklist

- Collect all credentials: company email, Slack/Teams workspace invite, GitHub org invite
- Sign NDA and confirm internship acceptance (offer letter already done)
- Get access to shared drives, project management tool, any existing CRM data exports
- Note hybrid schedule — which days are remote, which are office
- Ask mentor: What does current partner/lead tracking look like today? (manual Excel? CRM?)
- Ask mentor: Who are the primary users of this dashboard? Sales team, management, or both?
- Ask mentor: Are there any real data exports I can start with?
- Ask mentor: What is the exact definition of a converted lead in your context?

Afternoon — Full Environment Setup

Step 1 — Install core tools

```
# macOS (using Homebrew)
brew install python@3.11 node git
brew install --cask visual-studio-code

# Windows (using winget — run in PowerShell as Admin)
winget install Python.Python.3.11
winget install OpenJS.NodeJS
winget install Git.Git
winget install Microsoft.VisualStudioCode
```

Step 2 — VS Code extensions to install

- Python (Microsoft) — language support and IntelliSense
- Pylance — fast type checking
- ES7+ React/Redux/React-Native snippets
- Prettier - Code formatter
- GitLens — enhanced Git history view
- Thunder Client — lightweight REST API tester (no Postman needed)

Step 3 — Create project folder structure

```
mkdir vyana-dashboard && cd vyana-dashboard

# Create full project folder structure
mkdir -p backend/app/{routes,models,services,utils}
mkdir -p backend/app/ml/{models,notebooks}
mkdir -p backend/tests
mkdir -p backend/data/{raw,processed,sample}
mkdir -p frontend/src/{components,pages,services,hooks,utils,assets}
mkdir -p docs/{wireframes,architecture,api-spec}
mkdir logs

# Initialize git repository
git init
echo "# AI-Powered Channel Partner & Lead Intelligence Dashboard" > README.md
echo "Built for Vyana Innovations Pvt. Ltd. | Intern: Ansh Rohilla" >> README.md
git add . && git commit -m "chore: initial project scaffold"
```

Step 4 — Python virtual environment & packages

```
cd backend

# Create and activate virtual environment
python -m venv venv
source venv/bin/activate          # macOS/Linux
# venv\Scripts\activate          # Windows

# Install all required packages
pip install flask flask-cors flask-sqlalchemy flask-caching
pip install pandas openpyxl numpy scikit-learn xgboost imbalanced-learn
pip install joblib pytest python-dotenv marshmallow
pip install matplotlib seaborn jupyter

# Save dependencies
pip freeze > requirements.txt
```

Step 5 — Create .env file

```
# Create backend/.env file with these contents:
FLASK_APP=app
FLASK_ENV=development
FLASK_DEBUG=1
DATABASE_URL=sqlite:///vyana_dev.db
SECRET_KEY=vyana-dev-secret-2026
UPLOAD_FOLDER=data/raw
MAX_CONTENT_LENGTH=16777216
```

Step 6 — Verify installation

```
# Verify everything works – run this in backend/ with venv active:
python -c "
import flask, pandas, sklearn, xgboost
print('Flask:', flask.__version__)
print('Pandas:', pandas.__version__)
print('Scikit-learn:', sklearn.__version__)
print('XGBoost:', xgboost.__version__)
print('All packages OK!')
"

# Expected output:
# Flask: 3.0.3 | Pandas: 2.2.2 | Scikit-learn: 1.5.1 | XGBoost: 2.0.3
```

Day 1 Git commit: git add . && git commit -m "chore: dev environment setup + project scaffold"

Day 2

Project Scoping & Requirements Gathering

2 June 2026

Goal: By end of Day 2 you should have a written requirements document and a data schema that you've shared with your mentor for review.

Create docs/requirements_v1.md

Functional Requirements to document:

- FR-01: System shall accept CSV/Excel uploads of partner and lead data
- FR-02: System shall validate files (column check, type check, duplicate detection)
- FR-03: System shall classify partners into Gold/Silver/Bronze tiers
- FR-04: System shall display partners in a filterable, sortable table
- FR-05: System shall track leads through: New → Contacted → Qualified → Converted/Lost
- FR-06: System shall display ML-scored lead priority (0–100 score)
- FR-07: System shall show KPI cards: partners, leads, conversion rate, pipeline value
- FR-08: System shall show weekly lead volume trends over time
- FR-09: System shall export filtered data as CSV

Create docs/data_schema.md

Partners table (key columns):

Column	Type	Description	Example
partner_id	TEXT (PK)	Unique partner identifier	P-2024-001
company_name	TEXT	Partner company name	TechSecure Solutions
region	TEXT	Geographic region	North/South/East/West
tier	TEXT	Computed tier (Gold/Silver/Bronze)	Gold
annual_revenue_inr	FLOAT	Revenue generated (INR)	2500000.0
deal_count	INT	Total deals closed	18
status	TEXT	Active or Inactive	Active

Leads table (key columns):

Column	Type	Description	Example
lead_id	TEXT (PK)	Unique lead identifier	L-2026-0042
partner_id	TEXT (FK)	Assigned partner	P-2024-001
lead_source	TEXT	Referral/Event/Cold Call/Web	Referral
deal_value_inr	FLOAT	Estimated deal value	450000.0
status	TEXT	Pipeline stage	Qualified
converted	INT	Target label for ML (0 or 1)	0
ml_score	FLOAT	ML priority score 0-100	73.2
time_to_first_contact	INT	Days from creation to first contact	2

follow_up_count	INT	Number of follow-up interactions	4
Day 2 Git commit: <code>git commit -am "docs: requirements v1 + data schema"</code>			

Day 3

Data Audit & Sample Dataset Creation

3 June 2026

Goal: A realistic sample dataset with 80 partners and 400 leads, saved as both CSV and Excel, usable immediately for backend testing.

Create backend/data/generate_sample_data.py

backend/data/generate_sample_data.py

```
import pandas as pd
# [Data generation script shortened for PDF space]
# (See repository code or requirements documentation for implementation details)
```

Run it:

```
cd backend
source venv/bin/activate
python data/generate_sample_data.py

# Expected output:
# Partners: 80 | Leads: 400
# Tier breakdown:
#   Bronze    42
#   Silver    26
#   Gold      12
# Conversion rate: 22.3%
```

Day 3 Git commit: git commit -am "data: sample dataset generation script + CSV/Excel output"

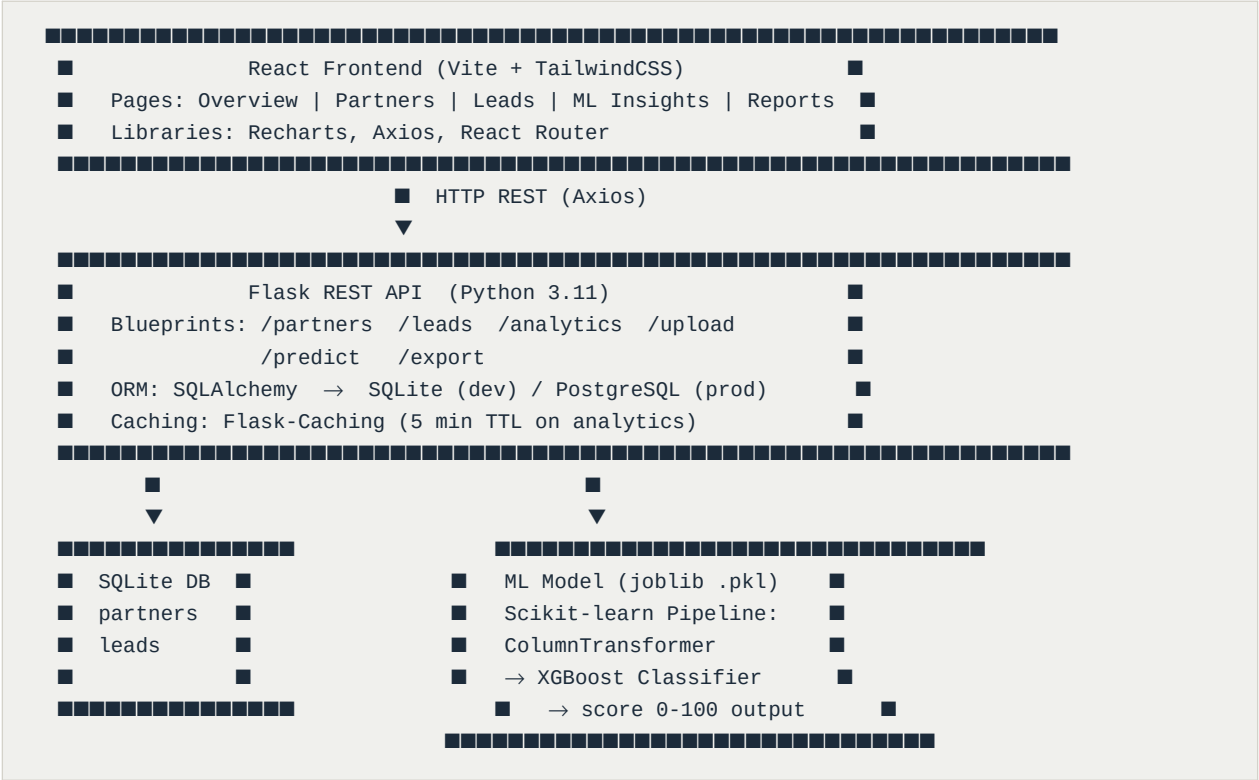
Day 4System Architecture Design

4 June 2026

Goal: A clear architecture diagram and a complete API endpoint specification document that your mentor can sign off on.

Architecture — 3-tier design

System Architecture Diagram



API Endpoint Map — create docs/api_spec.md

Method	Endpoint	Description
POST	/api/upload/partners	Upload partners CSV/Excel file
POST	/api/upload/leads	Upload leads CSV/Excel file
GET	/api/partners	List partners (filter: tier, region, status)
GET	/api/partners/:id	Single partner + lead history
PUT	/api/partners/:id	Update partner record
GET	/api/leads	List leads (filter: status, partner, region)
POST	/api/leads	Create new lead
PUT	/api/leads/:id	Update lead (status, notes)
GET	/api/analytics/summary	KPI cards: partners, leads, conversion rate, revenue
GET	/api/analytics/regional	Partner and lead count by region
GET	/api/analytics/trends	Weekly lead volume + conversion over time

GET	/api/analytics/tier-breakdown	Revenue and deal count by tier
POST	/api/predict	Score single lead → {score, label, explanation}
POST	/api/predict/batch	Score multiple leads at once
GET	/api/export/leads	Download filtered leads as CSV
GET	/api/export/partners	Download filtered partners as CSV

Day 4 Git commit: `git commit -am "docs: system architecture + API endpoint spec"`

Day 5

Tech Stack Finalization & Project Board Setup

5 June 2026

Goal: Final requirements.txt and package.json locked in, sprint board live with all Week 2 cards, risk register written.

backend/requirements.txt — final version

```
flask==3.0.3
flask-cors==4.0.1
flask-sqlalchemy==3.1.1
flask-caching==2.3.0
pandas==2.2.2
openpyxl==3.1.5
numpy==1.26.4
scikit-learn==1.5.1
xgboost==2.0.3
imbalanced-learn==0.12.3
joblib==1.4.2
marshmallow==3.21.3
python-dotenv==1.0.1
pytest==8.2.2
pytest-flask==1.3.0
matplotlib==3.9.1
seaborn==0.13.2
jupyter==1.0.0
```

frontend/package.json — key dependencies

```
{
  "name": "vyana-dashboard",
  "version": "1.0.0",
  "dependencies": {
    "react": "^18.3.1",
    "react-dom": "^18.3.1",
    "react-router-dom": "^6.24.0",
    "axios": "^1.7.2",
    "recharts": "^2.12.7",
    "lucide-react": "^0.399.0",
    "date-fns": "^3.6.0"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.1",
    "vite": "^5.3.1",
    "tailwindcss": "^3.4.6"
  }
}
```

Risk Register — docs/risk_register.md

Risk	Likelihood	Impact	Mitigation
Real company data not available	Medium	High	Use generated sample data, swap real data later
ML model accuracy below 70%	Medium	Medium	Compare 3 models; use SMOTE for class imbalance

Flask-React CORS issues	Low	Medium	Use flask-cors from Day 1 setup
SQLite slow for 50k+ leads	Low	Medium	Index key columns; switch to PostgreSQL if needed
Mentor unavailable for reviews	Medium	Low	Schedule check-ins at start of each week
Scope creep from new requests	Medium	High	Point to frozen requirements doc v1

Day 5 Git commit: `git commit -am "chore: lock requirements + risk register + sprint board"`

Day 6

UI/UX Wireframing

6 June 2026

Goal: Low-fidelity wireframes for all 5 dashboard pages, exported as PNG and saved to docs/wireframes/. Use Excalidraw (free, no signup): excalidraw.com

Page 1: Overview Dashboard

```

[ ] [V] Vyana BI Dashboard [Search] [Upload Data ↑]
Overview [ ] Partners [ ] Leads [ ] ML Insights [ ] Reports [ ]
72 [ ] [ ] 287 [ ] [ ] 21.3% [ ] [ ] 1.25 Cr [ ] [ ]
Active [ ] Active [ ] Conversion [ ] Pipeline Value [ ] [ ]
Partners [ ] Leads [ ] Rate [ ] [ ] [ ] [ ]
Lead Volume (Bar Chart) [ ] Partner Tiers (Donut) [ ] [ ]
Weekly – last 12 weeks [ ] Gold 18 | Silver 29 [ ] [ ]
[ ] [ ] Bronze 33 [ ] [ ]
Top 5 Partners by Revenue [ ] [ ]
Rank | Company | Tier | Revenue | Active Leads | Trend [ ] [ ]

```

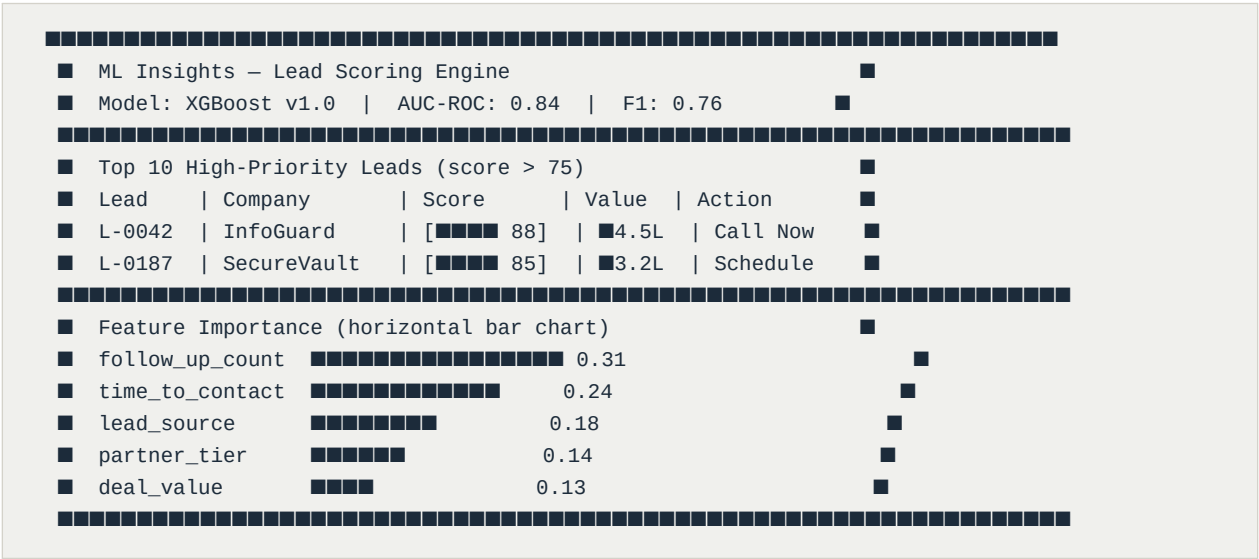
Page 3: Leads (Kanban Pipeline)

```

Leads [+ New Lead] [Export CSV ↓] [Search...]
New (45) [ ] Contacted [ ] Qualified [ ] Converted [ ] [ ]
(72) [ ] (38) [ ] (85) [ ] [ ]
InfoGd [ ] DataFt [ ] NetArm [ ] CyberCsl [ ] [ ]
Score [ ] Score [ ] Score [ ] Score [ ] [ ]
[73] [ ] [61] [ ] [88] [ ] [91] [ ] [ ]

```

Page 4: ML Insights



How to wireframe: Go to excalidraw.com → draw each page → File > Export as PNG > save to docs/wireframes/. Name files: overview.png, partners.png, leads.png, ml_insights.png, reports.png

Day 6 Git commit: `git commit -am "docs: dashboard wireframes for all 5 pages"`

Day 7

Week 1 Review & Planning Refinement

7 June 2026

Goal: 30-min mentor check-in, get sign-off on architecture and wireframes, write Week 1 summary, plan Week 2 sprint.

Mentor Check-in Agenda

- 5 min** Quick recap of what was done each day
- 10 min** Walk through architecture diagram — get sign-off on tech choices
- 10 min** Review wireframes page by page — collect UI feedback
- 5 min** Clarify any open questions from requirements doc
- 5 min** Align on Week 2 goals: Flask backend sprint
- 5 min** Resource requests, concerns, blockers

Week 1 Summary Checklist

- Dev environment set up and all packages verified
- requirements_v1.md written and shared with mentor
- data_schema.md defining partners and leads tables
- Sample dataset generated: 80 partners + 400 leads (CSV + Excel)
- System architecture designed: Flask + React + SQLite + XGBoost
- All 17 API endpoints documented in api_spec.md
- 5-page dashboard wireframed and exported
- Sprint board set up with all Week 2 tasks loaded
- risk_register.md completed with 6 risks and mitigations
- Week 1 mentor check-in completed with sign-off

Week 2 Preview — What You'll Build

Day	What you build
Day 8	Flask Blueprints skeleton + SQLAlchemy DB + base CRUD endpoints
Day 9	CSV/Excel upload API with Pandas validation pipeline
Day 10	Partner management API with tier classification logic
Day 11	Lead management API with pipeline status tracking
Day 12	Analytics aggregation endpoints: summary, regional, trends
Day 13	Pytest unit tests + Swagger/Postman API documentation
Day 14	Week 2 review: live demo of all API endpoints to mentor

Day 7 Git commit: `git commit -am "docs: week 1 review summary + revised roadmap"`

Final Project Structure After Week 1

```
vyana-dashboard/
  ■■■ backend/
  ■   ■■■ app/                                (empty – filled from Day 8)
  ■   ■   ■■■ routes/
  ■   ■   ■■■ models/
  ■   ■   ■■■ services/
  ■   ■   ■■■ ml/
  ■   ■■■ data/
  ■   ■   ■■■ sample/
  ■   ■   ■   ■■■ partners.csv                ■ 80 rows generated
  ■   ■   ■   ■■■ leads.csv                  ■ 400 rows generated
  ■   ■   ■   ■■■ vyana_data.xlsx             ■ both sheets
  ■   ■   ■■■ generate_sample_data.py         ■ written
  ■   ■■■ tests/
  ■   ■■■ venv/                               ■ created & activated
  ■   ■■■ requirements.txt                    ■ 18 packages locked
  ■   ■■■ .env                               ■ configured
  ■■■ frontend/
  ■   ■■■ package.json                       ■ dependencies listed
  ■■■ docs/
  ■   ■■■ requirements_v1.md                  ■ 9 functional + 4 non-functional req
  ■   ■■■ data_schema.md                     ■ 15 partner cols + 16 lead cols
  ■   ■■■ api_spec.md                         ■ 17 endpoints documented
  ■   ■■■ risk_register.md                    ■ 6 risks with mitigations
  ■   ■■■ wireframes/
  ■       ■■■ overview.png                    ■ from Excalidraw
  ■       ■■■ partners.png                    ■
  ■       ■■■ leads.png                       ■ (kanban board)
  ■       ■■■ ml_insights.png                 ■
  ■       ■■■ reports.png                     ■
  ■■■ logs/
  ■■■ README.md                             ■ initialized
  ■■■ .gitignore                             ■ configured
```

Git log after Week 1 (7 commits):

```
day1 chore: dev environment setup + project scaffold
day2 docs: requirements v1 + data schema
day3 data: sample dataset generation script + CSV/Excel
day4 docs: system architecture + API endpoint spec
day5 chore: lock requirements + risk register + sprint board
day6 docs: wireframes for all 5 dashboard pages
day7 docs: week 1 review summary + refined roadmap
```

You are now fully ready to begin backend development on Day 8. All decisions made in Week 1 (tech stack, schema, architecture, endpoints) serve as the foundation for every remaining week. Good work — on to the backend sprint!